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EE301

Power Generation, Transmission And Protection

MODULE I – PART 1

Thermal Power Plant

KTU Syllabus - Semester 5

**Department of Electrical and Electronics
Engineering**

Engineering, Irinjalakuda

MODULE I

Introduction: Typical layout of Power system network Generation of Electric Power

Economics of Generation

Module	Contents	Hours	Sem. Exam Marks
I	Introduction: Typical layout of Power system Network Generation of Electric Power: Overview of conventional (Hydro, Thermal and Nuclear) and Nonconventional Sources (Solar and Wind) (Block Diagram and Brief Description Only) Economics of Generation: Load factor, diversity factor, Load curve (Brief description only) Numerical Problems. Methods of power factor improvement using capacitors	9	15%



GENERATING STATIONS

- Bulk Electric power is produced by special plants known as generating stations or power plants.
- A generating station has a prime mover coupled to an alternator for the production of electric power. The prime mover (e.g., steam turbine, water turbine etc.) converts energy from some other form into mechanical energy. The alternator converts mechanical energy of the prime mover into electrical energy.



Classification of Generating Stations

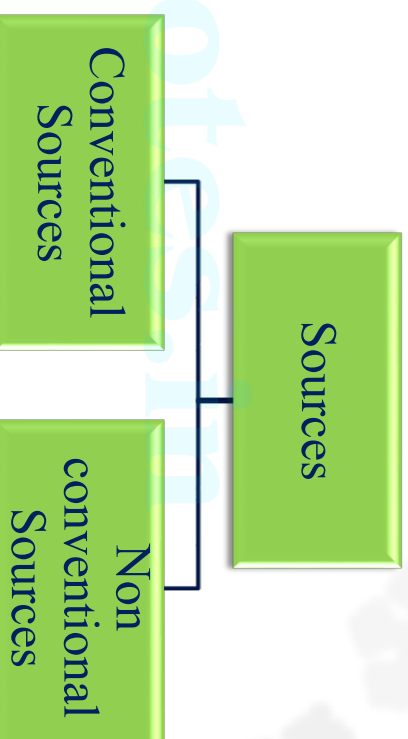
Depending upon the form of energy converted into electrical energy, the generating stations are classified as under :

Conventional Sources

- (i) Thermal / Steam power stations
- (ii) Hydroelectric power stations
- (iii) Nuclear power stations

Non-Conventional Sources

- (i) Solar Power Station
- (ii) Wind Power Plant



Conventional Sources

Thermal or Steam Power Plant

- *A generating station which converts heat energy of coal combustion into electrical energy is known as a **steam power station**.*
- A thermal power plant employs a prime mover (steam turbine) coupled to an alternator for the production of electric power.
- The steam turbine converts heat energy into mechanical energy. The alternator is coupled to the steam turbine.
- The alternator converts mechanical energy to electrical energy. The electrical energy produced by the generating station is transmitted.

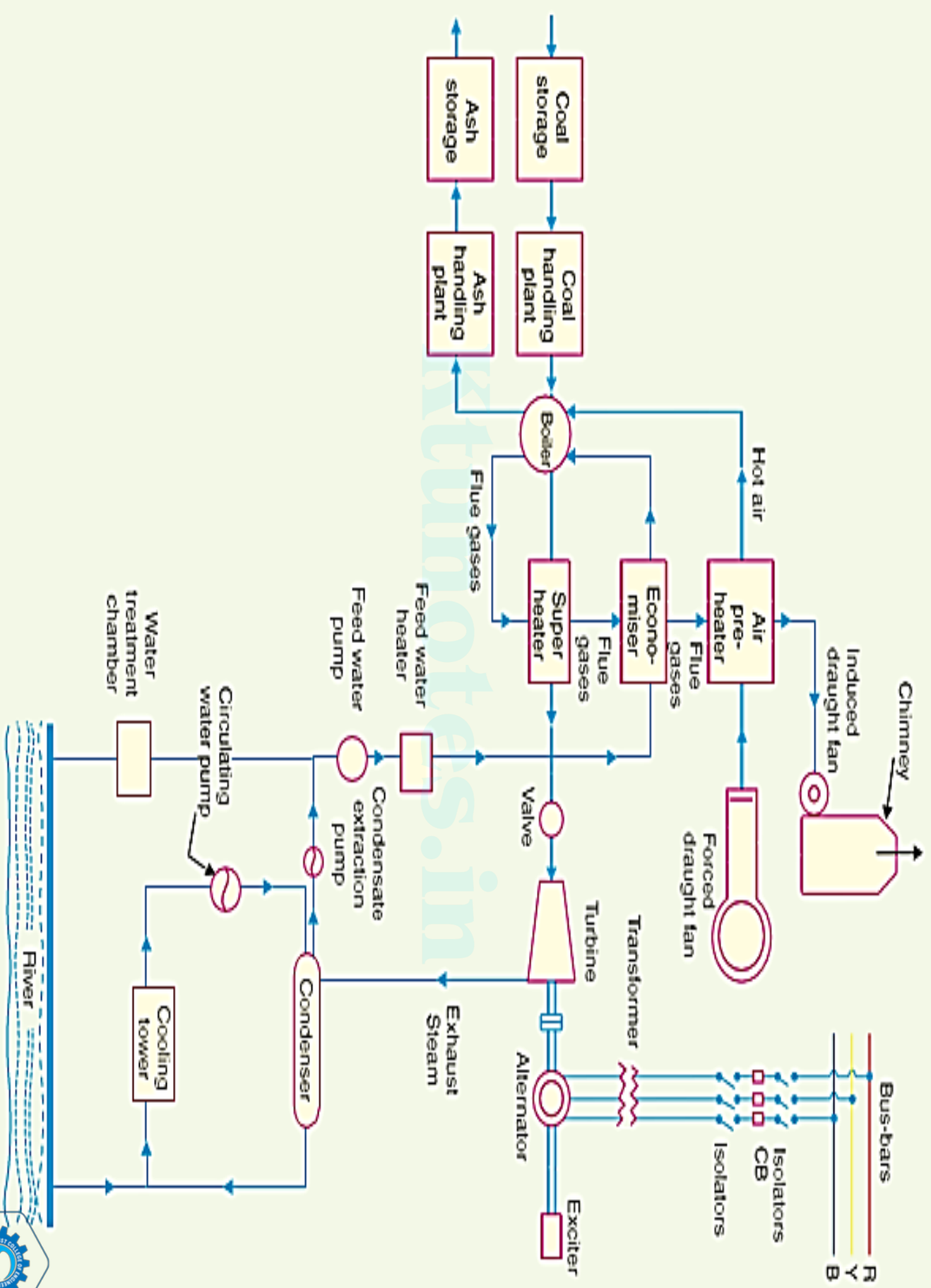


Schematic Arrangement of Steam Power Station

1. Coal and ash handling arrangement
2. Steam generating plant
3. Steam turbine
4. Alternator
5. Feed water
6. Cooling arrangement

1. Coal and ash handling plant. The coal is transported to the power station by road or rail and is stored in the coal storage plant. From the coal storage plant, coal is delivered to the coal handling plant where it is pulverised (*i.e.*, crushed into small pieces). The pulverised coal is fed to the boiler by belt conveyors. The coal is burnt in the boiler and the ash is removed to the ash handling plant and then delivered to the ash storage plant for disposal.





2. Boiler. The coal is burnt in the boiler and the heat is utilised to convert water into steam at high temperature and pressure. The flue gases from the boiler is passed through super-heater, economiser, air pre-heater and are finally exhausted to atmosphere through the chimney.

3. Superheater. The steam produced in the boiler is wet and is passed through a superheater where it is dried and superheated. Superheater improves the overall efficiency and prevents corrosion of the turbine blades. The superheated steam from the superheater is fed to steam turbine through the main valve.

4. Economiser. An economiser is essentially a feed water heater and derives heat from the flue gases. The feed water is fed to the economiser before supplying to the boiler.

5. Air preheater. An air preheater increases the temperature of the air supplied for coal burning by deriving heat from flue gases. Air is drawn from the atmosphere by a forced draught fan and is passed through air preheater before supplying to the boiler furnace.



6. Steam turbine. The dry and superheated steam from the superheater is fed to the steam turbine through main valve. The heat energy of steam when passing over the blades of turbine is converted into mechanical energy. The steam is exhausted to the *condenser* which condenses the exhausted steam by means of cold water circulation.

7. Alternator. The steam turbine is coupled to an alternator. The alternator converts mechanical energy of turbine into electrical energy. The electrical output from the alternator is delivered to the bus bars through transformer, circuit breakers and isolators.

8. Feed water. The condensate from the condenser is used as feed water to the boiler. Some water may be lost in the cycle which is suitably filled up from external source.

9. Cooling arrangement. In order to improve the efficiency of the plant, the steam exhausted from the turbine is condensed by means of a condenser. Water is drawn from a natural source of supply such as a river, canal or lake and is circulated through the condenser. The circulating water takes up the heat of the exhausted steam and itself becomes hot. This hot water coming out from the condenser is discharged at a suitable location down the river.



Advantages

- The fuel (*i.e.*, coal) used is quite cheap.
- Less initial cost as compared to other generating stations.
- It can be installed at any place irrespective of the existence of coal.
- The coal can be trans-ported to the site of the plant by rail or road.
- It requires less space as compared to the hydroelectric power station.
- The cost of generation is lesser than that of the diesel power station.

Disadvantages

- It pollutes the atmosphere due to the production of large amount of smoke and fumes.
- It is costlier in running cost as compared to hydroelectric plant.

HYDROELECTRIC POWER PLANT

- *A generating station which utilises the potential energy of water at a high level for the generation of electrical energy is known as a **hydro-electric power station**.*

- In a hydro-electric power station, water head is created by constructing a dam across a river or lake. From the dam, water is led to a water turbine. The water turbine captures the energy in the falling water and changes the hydraulic energy into mechanical energy at the turbine shaft. The turbine drives the alternator which converts mechanical energy into electrical energy.



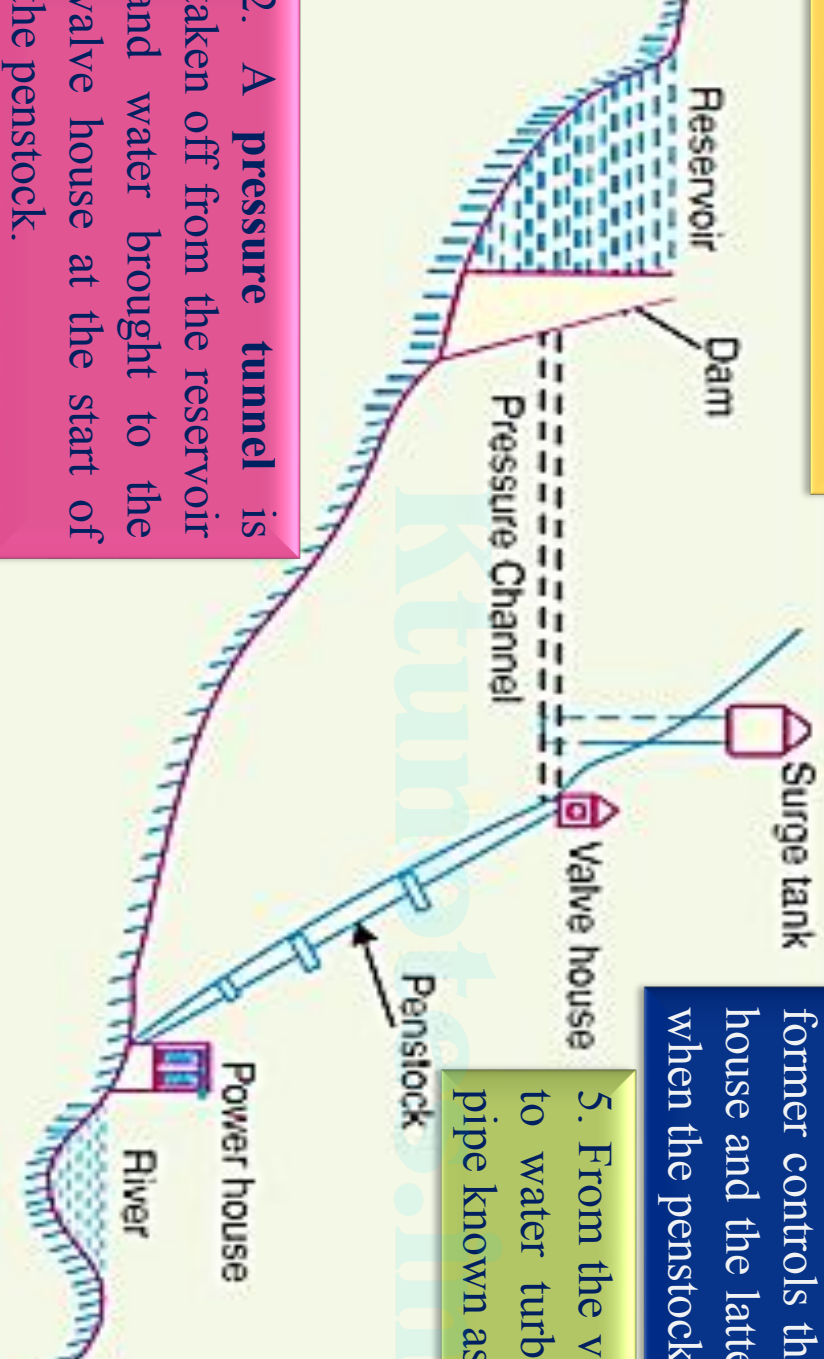
1. A **dam** is constructed across a river or lake and water from the catchment area collects at the back of the dam to form a reservoir.

3. A **surge tank** (open from top) is built just before the valve house and protects the penstock from bursting in case the turbine gates suddenly close due to electrical load being thrown off.

4. The **valve house** contains main sluice valves and automatic isolating valves. The former controls the water flow to the power house and the latter cuts off supply of water when the penstock bursts

5. From the valve house, water is taken to water turbine through a huge steel pipe known as **penstock**.

6. In the **power house**, the water turbine converts hydraulic energy into mechanical energy. The turbine drives the alternator which converts mechanical energy into electrical energy.



Schematic arrangement of a Hydro-electric plant

2. A **pressure tunnel** is taken off from the reservoir and water brought to the valve house at the start of the penstock.

Advantages

- It requires no fuel as water is used for the generation of electrical energy.
- It is quite neat and clean as no smoke or ash is produced.
- It requires very small running charges because water is the source of energy which is available free of cost.
- It is comparatively simple in construction and requires less maintenance.
- It does not require a long starting time like a steam power station. In fact, such plants can be put into service instantly.
- It is robust and has a longer life.
- Such plants serve many purposes. In addition to the generation of electrical energy, they also help in irrigation and controlling floods.
- Although such plants require the attention of highly skilled persons at the time of construction, yet for operation, a few experienced persons may do the job well.

Disadvantages

- It involves high capital cost due to construction of dam.
- There is uncertainty about the availability of huge amount of water due to dependence on weather conditions.
- Skilled and experienced hands are required to build the plant.
- It requires high cost of transmission lines as the plant is located in hilly areas which are quite away from the consumers.

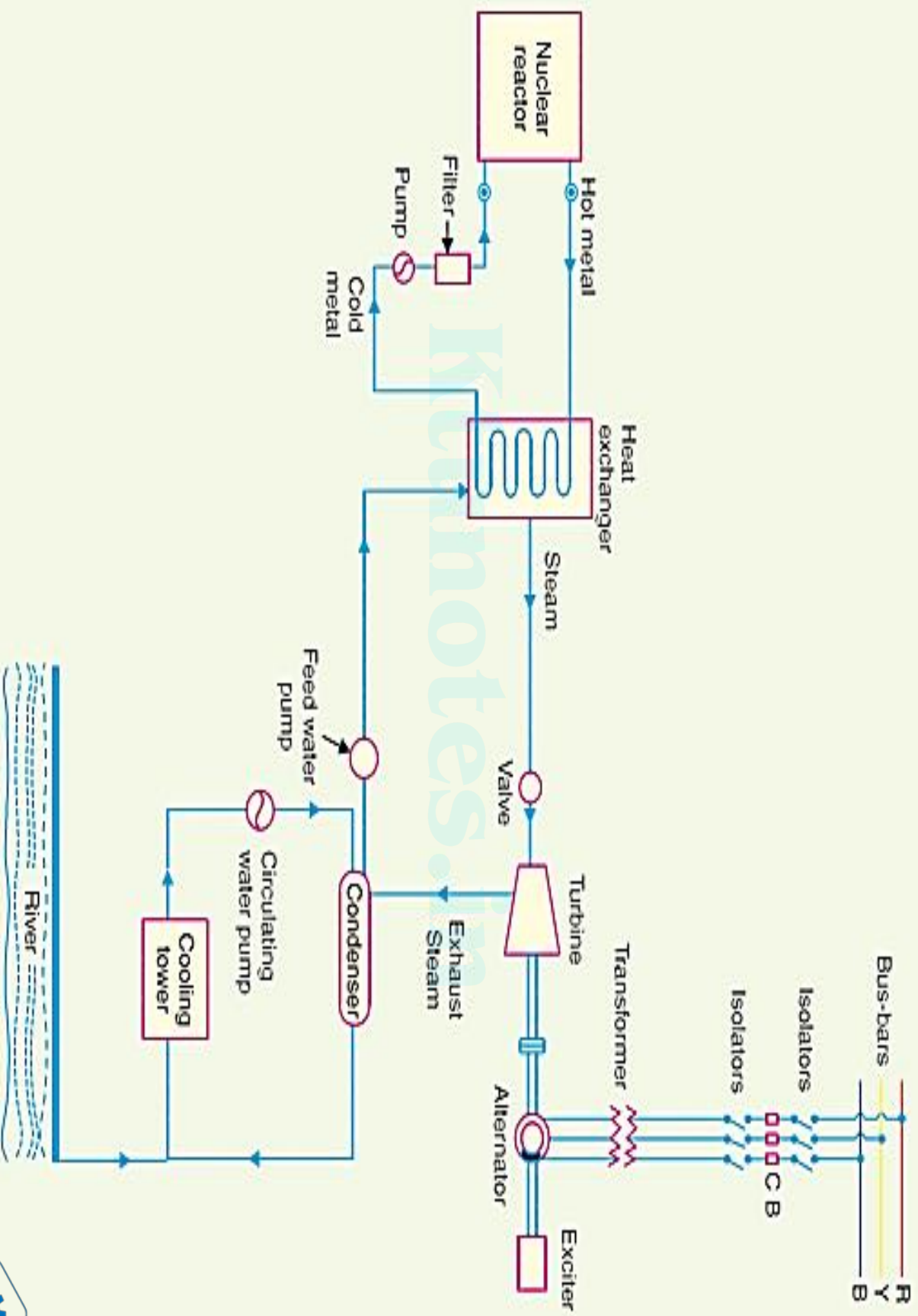


NUCLEAR POWER PLANT

- *A generating station in which nuclear energy is converted into electrical energy is known as a nuclear power station.*
- In nuclear power station, heavy elements such as Uranium (U^{235}) or Thorium (Th^{232}) are subjected to nuclear fission* in a special apparatus known as a *reactor*. The heat energy thus released is utilised in raising steam at high temperature and pressure. The steam runs the steam turbine which converts steam energy into mechanical energy. The turbine drives the alternator which converts mechanical energy into electrical energy.



Schematic Representation of Nuclear power plant

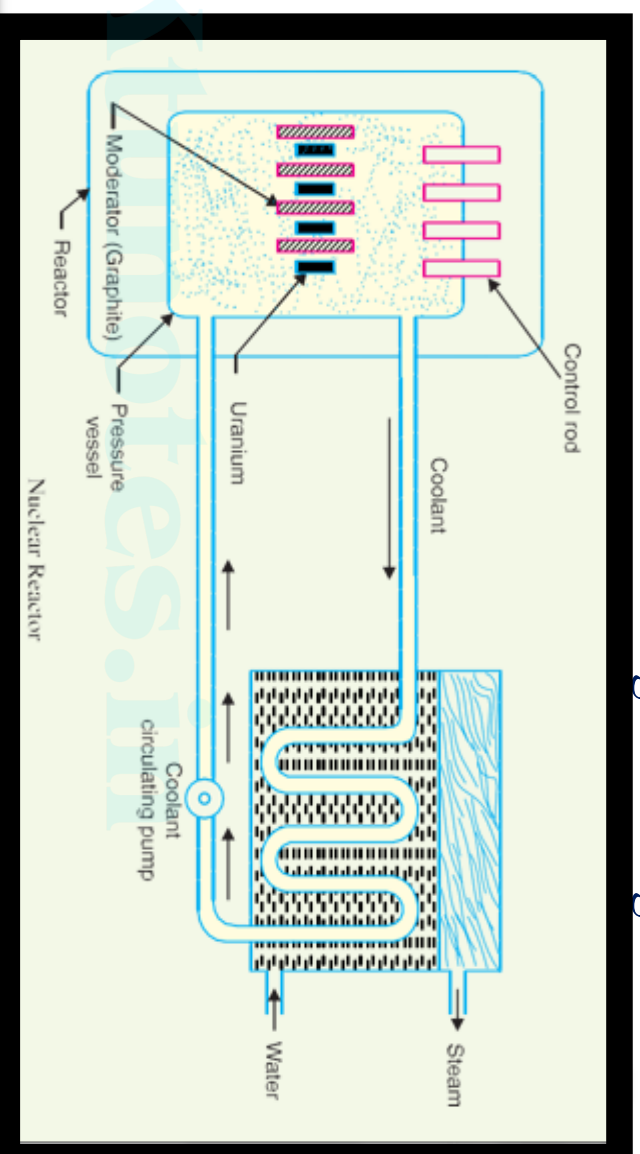


Schematic arrangement of Nuclear Power Station



- The schematic arrangement of a nuclear power station is shown in Fig. The whole arrangement can be divided into the following main stages :

1. Nuclear reactor
2. Heat exchanger
3. Steam turbine
4. Alternator.



1. Nuclear reactor. It is an apparatus in which nuclear fuel (U^{235}) is subjected to nuclear fission. It controls the *chain reaction* that starts once the fission is done. If the chain reaction is not controlled, the result will be an explosion due to the fast increase in the energy released.

2. Heat exchanger. The coolant gives up heat to the heat exchanger which is utilised in raising the steam. After giving up heat, the coolant is again fed to the reactor.

3. Steam turbine. The steam produced in the heat exchanger is led to the steam turbine through a valve. After doing a useful work in the turbine, the steam is exhausted to condenser. The condenser condenses the steam which is fed to the heat exchanger through feed water pump

4. Alternator. The steam turbine drives the alternator which converts mechanical energy into electrical energy. The output from the alternator is delivered to the bus-bars through trans- former, circuit breakers and isolators.



Advantages

- The amount of fuel required is quite small. Therefore, there is a considerable saving in the cost of fuel transportation.
- A nuclear power plant requires less space as compared to any other type of the same size.
- It has low running charges as a small amount of fuel is used for producing bulk electrical energy.
- This type of plant is very economical for producing bulk electric power.
- It can be located near the load centres because it does not require large quantities of water and need not be near coal mines. Therefore, the cost of primary distribution is reduced.
- There are large deposits of nuclear fuels available all over the world. Therefore, such plants can ensure continued supply of electrical energy for thousands of years.
- It ensures reliability of operation.

Disadvantages

- The fuel used is expensive and is difficult to recover.
- The capital cost on a nuclear plant is very high as compared to other types of plants.
- The erection and commissioning of the plant requires greater technical know-how.
- The fission by-products are generally radioactive and may cause a dangerous amount of radioactive pollution.

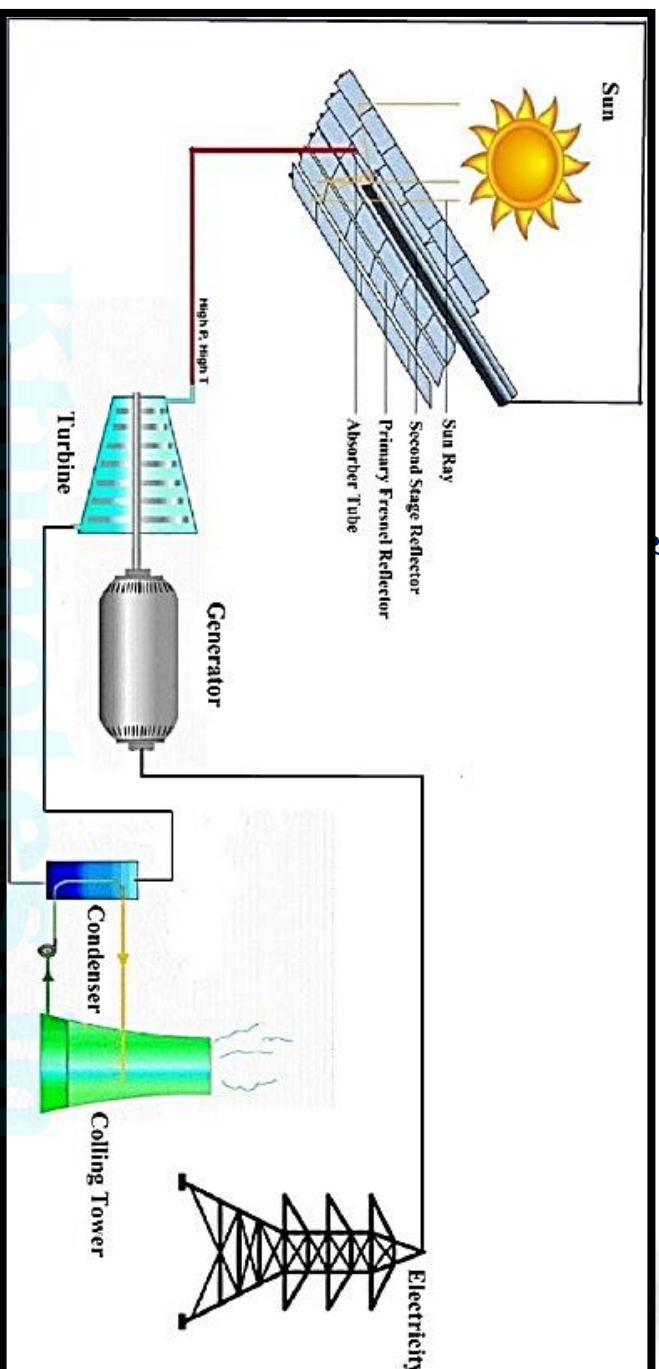
Non – Conventional Sources

SOLAR POWER PLANT

- The total solar energy reaching the earth is 1.77×10^{18} kWh/year.
- Solar radiation includes visible light, infrared rays and ultraviolet rays. Visible light and infrared (available as heat) rays are utilized for generation of electricity.
- There are 2 types of solar power plants:
 1. Solar Thermal systems
 2. Solar photovoltaic systems



Solar thermal system



- The basic principle is to concentrate the solar radiation to a receiver using mirror reflectors and the resulting heat is used to generate electricity. The conversion of heat energy to mechanical energy is done using steam turbine.

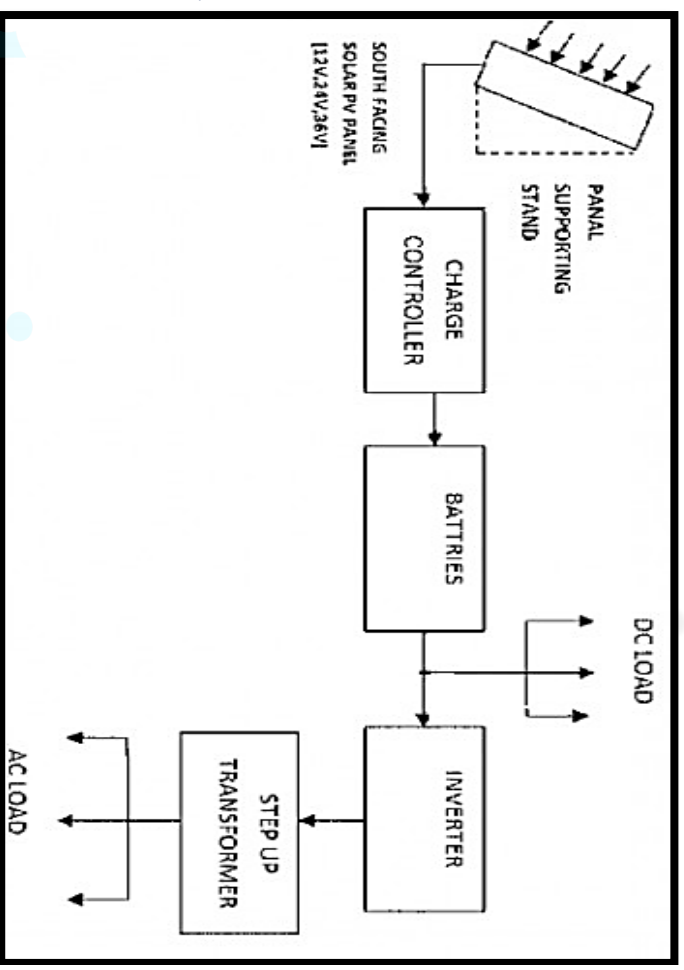
- Parabolic solar collector is commonly used. The parabolic disc reflector concentrates the solar radiation to a receiver. A large array of solar collectors is involved in power plants.

The heat produced is used to generate steam and in turn drive a turbine and generator to produce electricity.



Solar Photovoltaic system

- In solar photovoltaic systems the visible light is directly converted into electricity by solar cells usually made of silicon.
- A solar cell is basically a PN Junction diode which generates an electron flow when visible light falls on it.
- The voltage generated by a typical solar cell is around 0.5V only and a large number of such cells are connected in series to get the required voltage.



- A set of cells connected in series and properly laminated is called solar module. One or more solar modules produce electricity and is stored in a battery(usually lead acid type) for night time use.
- There is a charge controller in the system which controls the charging and discharging of the battery.
- This is to protect the battery from over charging and discharging.
- The power available from the solar module is in DC only and therefore dc loads can be directly connected if the voltage ratings of load and source are same.
- For AC loads, a DC-AC converter is needed.

SOLAR POWER PLANT

Advantages

1. Solar energy is renewable source and hence no fuel is required.
2. There is no atmospheric pollution.
3. There is no moving part so maintenance requirement is less.
4. Solar modules have long life.

Disadvantages

1. High cost of the system.
2. Energy generation only possible when sunlight is available.
3. Output is DC so an DC AC converter is needed.



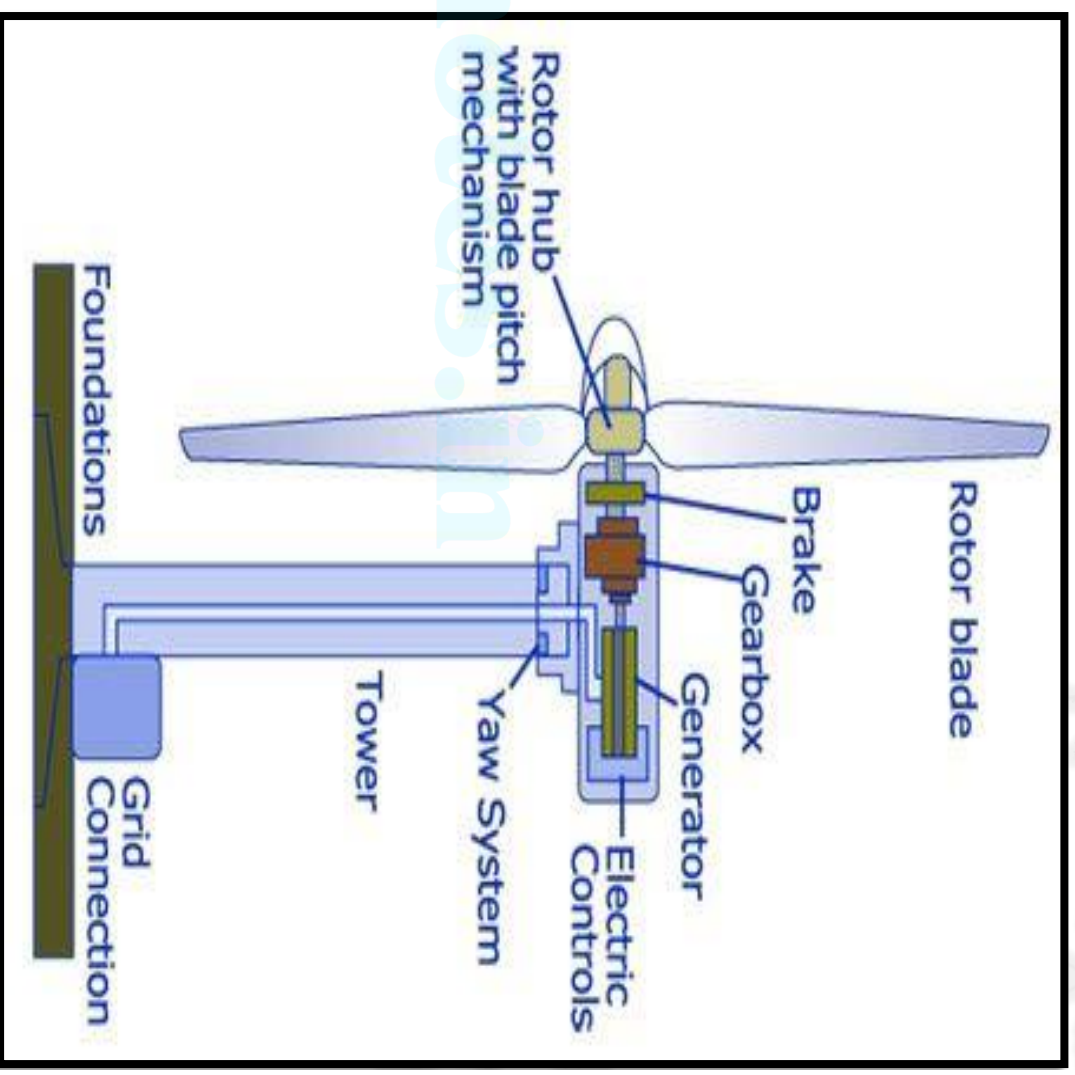
Wind Power plant

Brake is used to cut-off when there is extremely high wind.

Gearbox it improve the speed ratio

Generator converts mechanical energy into electrical energy.

Yaw system aligns the rotor blades according to the orientation of wind.



- A simplified illustration of a wind generator is shown in figure.
- A wind generator mainly consists of a wind turbine usually rotating in a horizontal axis, a generator, a speed changing gear system which couples the turbine with the generator and control equipment.
- The turbine and generator assembly is mounted on a tower of about 20-25 m height.
- Regions where steady wind in the range of 5 to 30 m/s is available are considered suitable for grid connected wind generators.
- An array of generators is installed in such locations.
- The individual capacity of generators is usually in the range of 250kW to 1 MW.

Wind Power plant

Advantages

1. Wind energy is renewable source and hence no fuel is required.
2. There is no atmospheric pollution.
3. Can be built on existing farms.

Disadvantages

1. High cost of the installation system.
2. Energy generation only possible when wind is available.
3. Can be installed on large open spaces.

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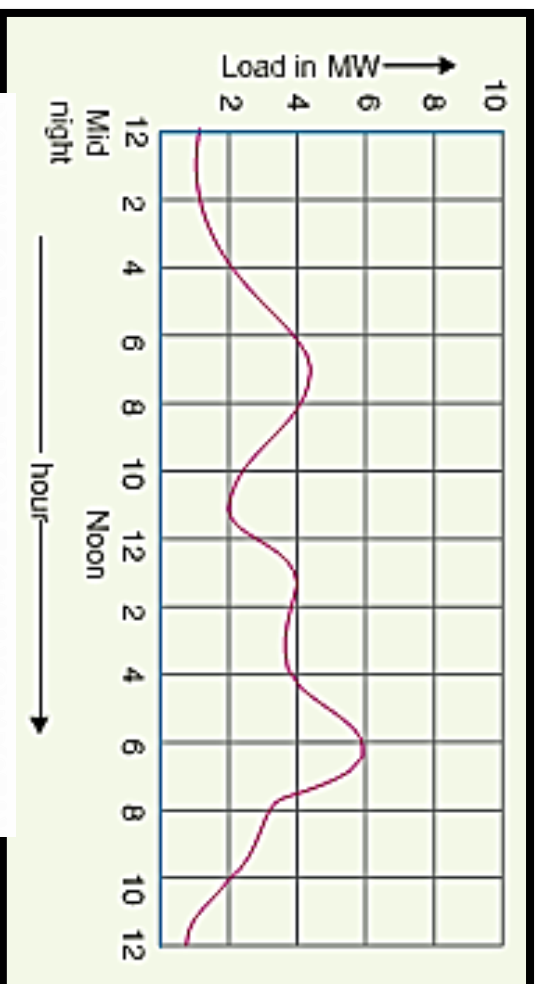


Load Curve

- The curve showing the variation of load on the power station with respect to (w.r.t) time is known as a **load curve**.

Daily Load Curve

- The load on a power station is never constant; it varies from time to time.
- These load variations during the whole day (*i.e.*, 24 hours) are recorded half-hourly or hourly and are plotted against time on the graph.
- The curve thus obtained is known as *daily load curve* as it shows the variations of load *w.r.t.* time during the day.
- Figure shows a typical daily load curve of a power station.



Monthly load curve

- The *monthly load curve* can be obtained from the daily load curves of that month.
- For this purpose, average values of power over a month at different times of the day are calculated and then plotted on the graph.

- The monthly load curve is generally used to fix the rates of energy.

Yearly load curve

- The *yearly load curve* is obtained by considering the monthly load curves of that particular year.
- The yearly load curve is generally used to determine the annual load factor.

Significance

- The daily load curve shows the variations of load on the power station during different hours of the day.
- The area under the daily load curve gives the number of units generated in the day.
- The highest point on the daily load curve represents the maximum demand on the station on that day.
- The area under the daily load curve divided by the total number of hours gives the average load on the station in the day.

$$\text{Average load} = \frac{\text{Area (in kWh) under daily load curve}}{24 \text{ hours}}$$

- The ratio of the area under the load curve to the total area of rectangle in which it is contained gives the load factor

$$\begin{aligned} \text{Load factor} &= \frac{\text{Average load}}{\text{Max. demand}} = \frac{\text{Average load} \times 24}{\text{Max. demand} \times 24} \\ &= \frac{\text{Area (in kWh) under daily load curve}}{\text{Total area of rectangle in which the load curve is contained}} \end{aligned}$$

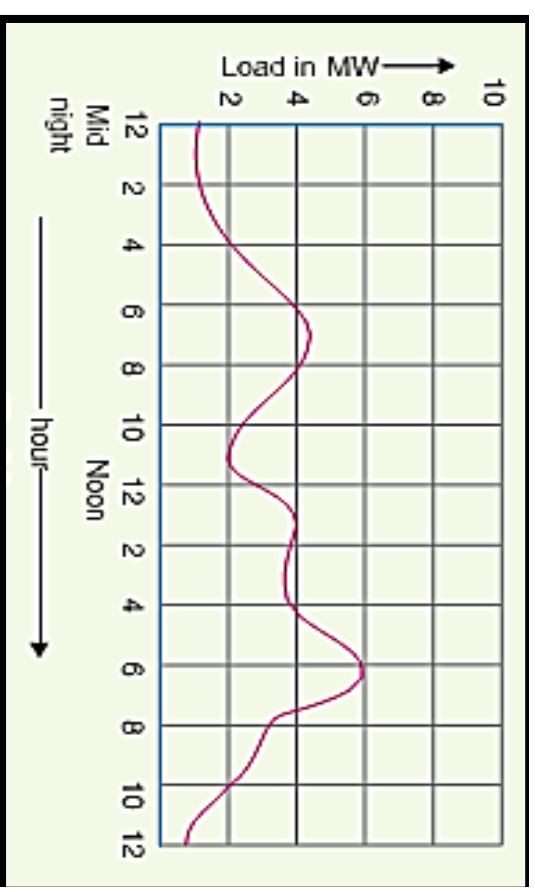
- The load curve helps in selecting the size and number of generating units
- The load curve helps in preparing the operation schedule of the station

1. Connected load

- It is the sum of continuous rating of all the equipments connected to supply system.
- For instance, if a consumer has connections of five 100-watt lamps and a power point of 500 watts, then connected load of the consumer is $5 \times 100 + 500 = 1000$ watts.

2. Maximum Demand

- It is the greatest demand of load on the power station during a given period.
- The maximum of all the demands that have occurred during a given period is the maximum demand.
- Thus referring back to the load curve of Fig the maximum demand on the power station during the day is 6 MW and it occurs at 6 P.M.
- Maximum demand is generally less than the connected load because all the consumers do not switch on their connected load to the system at a time.
- The knowledge of maximum demand is very important as it helps in determining the installed capacity of the station. The
- station must be capable of meeting the maximum demand.



3. Demand factor

- It is the ratio of maximum demand on the power station to its connected load.

$$\text{Demand factor} = \frac{\text{Maximum demand}}{\text{Connected load}}$$

- The value of demand factor is usually less than 1.
- It is expected because maximum demand on the power station is generally less than the connected load.
- If the maximum demand on the power station is 80 MW and the connected load is 100 MW, then demand factor = $80/100 = 0.8$.
- The knowledge of demand factor is vital in determining the capacity of the plant equipment.

4. Average load

- The average of loads occurring on the power station in a given period (day or month or year) is known as **average load** or **average**.

$$\begin{aligned}\text{Daily average load} &= \frac{\text{No. of units (kWh) generated in a day}}{24 \text{ hours}} \\ \text{Monthly average load} &= \frac{\text{No. of units (kWh) generated in a month}}{\text{Number of hours in a month}} \\ \text{Yearly average load} &= \frac{\text{No. of units (kWh) generated in a year}}{8760 \text{ hours}}\end{aligned}$$



5. Load factor

- The ratio of average load to the maximum demand during a given period is known as **load factor** .

$$\begin{aligned}\text{Load factor} &= \frac{\text{Average load}}{\text{Max. demand}} \\ \text{If the plant is in operation for } T \text{ hours,} \\ \text{Load factor} &= \frac{\text{Average load} \times T}{\text{Max. demand} \times T} \\ &= \frac{\text{Units generated in } T \text{ hours}}{\text{Max. demand} \times T \text{ hours}}\end{aligned}$$

- The load factor may be daily load factor, monthly load factor or annual load factor if the time period considered is a day or month or year.
- Load factor is always less than 1 because average load is smaller than the maximum demand.
- The load factor plays key role in determining the overall cost per unit generated.
- Higher the load factor of the power station, lesser will be the cost per unit generated.

6. Diversity Factor.

- The ratio of the sum of individual maximum demands to the maximum demand on power station is known as **diversity factor**.

$$\text{Diversity factor} = \frac{\text{Sum of individual max. Demands}}{\text{Max. demand on power station}}$$

- A power station supplies load to various types of consumers whose maximum demands generally do not occur at the same time.
- Therefore, the maximum demand on the power station is always less than the sum of individual maximum demands of the consumers.
- Obviously, diversity factor will always be greater than 1.
- The greater the diversity factor, the lesser is the cost of generation of power.

7. Plant capacity factor

- It is the ratio of actual energy produced to the maximum possible energy that could have been produced during a given period .

$$\begin{aligned}\text{Plant capacity factor} &= \frac{\text{Actual energy produced}}{\text{Max. energy that could have been produced}} \\ &= \frac{\text{Average demand} \times T}{\text{Plant capacity} \times T} \\ &= \frac{\text{Average demand}}{\text{Plant capacity}}\end{aligned}$$

- Thus if the considered period is one year

$$\text{Annual plant capacity factor} = \frac{\text{Annual kWh output}}{\text{Plant capacity} \times 8760}$$

- The plant capacity factor is an indication of the reserve capacity of the plant. A power station is so designed that it has some reserve capacity for meeting the increased load demand in future. Therefore, the installed capacity of the plant is always somewhat greater than the maximum demand on the plant.

$$\text{Reserve capacity} = \text{Plant capacity} - \text{Max. demand}$$

- It is interesting to note that difference between load factor and plant capacity factor is an indication of reserve capacity. If the maximum demand on the plant is equal to the plant capacity, then load factor and plant capacity factor will have the same value. In such a case, the plant will have no reserve capacity.

8. Plant use factor

- It is ratio of kWh generated to the product of plant capacity and the number of hours for which the plant was in operation.

$$\text{Plant use factor} = \frac{\text{Station output in kWh}}{\text{Plant capacity} \times \text{Hours of use}}$$

- Suppose a plant having installed capacity of 20 MW produces annual output of 7.35×10^6 kWh and remains in operation for 2190 hours in a year. Then,

$$\text{Plant use factor} = \frac{7.35 \times 10^6}{20 \times 10^3 \times 2190} = 0.167 = 16.7\%$$

Units Generated per Annum

- It is often required to find the kWh generated per annum from maximum demand and load factor. The procedure is as follows

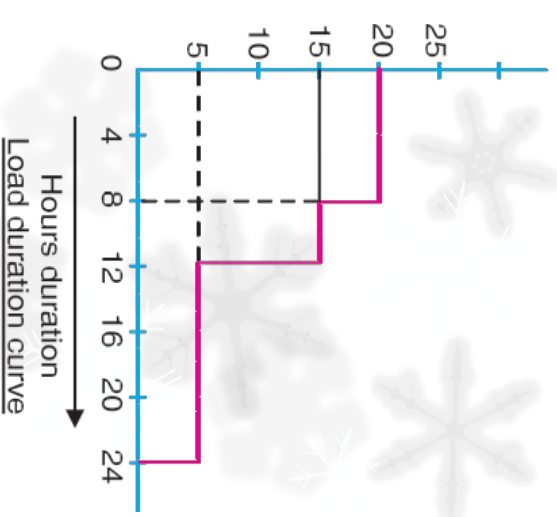
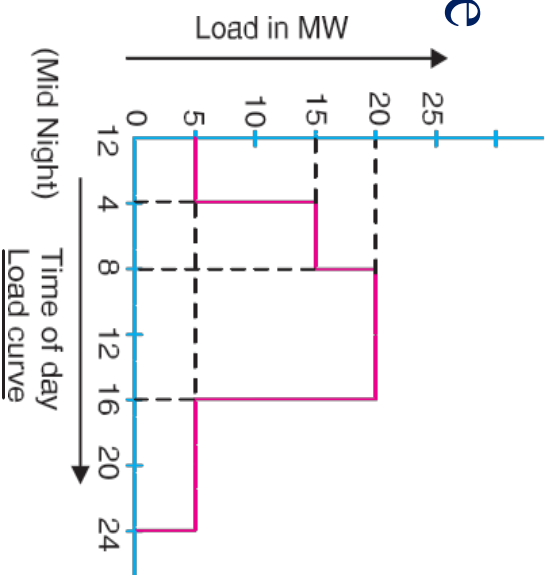
$$\begin{aligned}\text{Load factor} &= \frac{\text{Average load}}{\text{Max. demand}} \\ \text{Average load} &= \text{Max. demand} \times \text{L.F.} \\ \text{Units generated/annum} &= \text{Average load (in kW)} \times \text{Hours in a year} \\ &= \text{Max. demand (in kW)} \times \text{L.F.} \times 8760\end{aligned}$$

Tutorial

- A generating station has a connected load of 43MW and a maximum demand of 20 MW; the units generated being 61.5×10^6 per annum. Calculate (i) the demand factor and (ii) load factor.*
- A generating station has a maximum demand of 25MW, a load factor of 60%, a plant capacity factor of 50% and a plant use factor of 72%. Find (i) the reserve capacity of the plant (ii) the daily energy produced and (iii) maximum energy that could be produced daily if the plant while running as per schedule, were fully loaded.*



Load Duration Curve



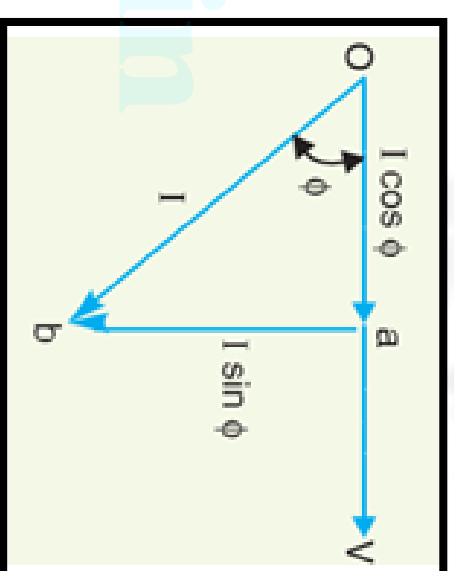
- When the load elements of a load curve are arranged in the order of descending magnitudes, the curve thus obtained is called a **load duration curve**.
- The load duration curve gives the data in a more presentable form. In other words, it readily shows the number of hours during which the given load has prevailed.
- The area under the load duration curve is equal to that of the corresponding load curve.
- Obviously, area under daily load duration curve (in kWh) will give the units generated on that day.
- The load duration curve can be extended to include any period of time.
- By laying out the abscissa from 0 hour to 8760 hours, the variation and distribution of demand for an entire year can be summarised in one curve.
- The curve thus obtained is called the *annual load duration curve*.

Power Factor Improvement

Power Factor

- The cosine of angle between voltage and current in an a.c. circuit is known as **power factor**.
- Consider an inductive circuit taking a lagging current I from supply voltage V ; the angle of lag being ϕ .

- The phasor diagram of the circuit is shown in Fig
- I can be resolved into 2 perpendicular components
- $I \cos \phi$ in phase with V
- $I \sin \phi$ 90° out of phase with V



- The component $I \cos \phi$ is known as active or wattful component, whereas component $I \sin \phi$ is called the reactive or wattless component.
- It may be noted that value of power factor can never be more than unity.
- Power factor, $\cos \phi = \frac{\text{Active Power}}{\text{Apparent Power}} = \frac{kW}{kVA}$

Disadvantages of low power Factor

- The power factor plays an importance role in a.c. circuits since power consumed depends upon this factor

$$\begin{aligned} P &= V_L I_L \cos \phi && \text{(For single phase supply)} \\ I_L &= \frac{P}{V_L \cos \phi} && \dots(i) \\ P &= \sqrt{3} V_L I_L \cos \phi && \text{(For 3 phase supply)} \\ I_L &= \frac{P}{\sqrt{3} V_L \cos \phi} && \dots(ii) \end{aligned}$$

- It is clear from above that for fixed power and voltage, the load current is inversely proportional to the power factor.
- Lower the power factor, higher is the load current and *vice-versa*.
- A power factor less than unity results in the following disadvantages :

Disadvantages of low power Factor

• **Large kVA rating of equipment.** The electrical machinery (e.g., alternators, transformers, switchgear) is always rated in kVA. It is clear that kVA rating of the equipment is inversely proportional to power factor. The smaller the power factor, the larger is the kVA rating. Therefore, at low power factor, the kVA rating of the equipment has to be made more, making the equipment larger and expensive.

$$\text{kVA} = \frac{\text{kW}}{\cos \phi}$$

• **Greater conductor size.** To transmit or distribute a fixed amount of power at constant voltage, the conductor will have to carry more current at low power factor. This necessitates large conductor size.

3. Large copper losses. The large current at low power factor causes more I^2R losses in all the elements of the supply system. This results in poor efficiency.

4. Poor voltage regulation. The large current at low lagging power factor causes greater voltage drops in alternators, transformers, transmission lines and distributors. This results in the decreased voltage available at the supply end, thus impairing the performance of utilisation devices. In order to keep the receiving end voltage within permissible limits, extra equipment (*i.e.*, voltage regulators) is required.

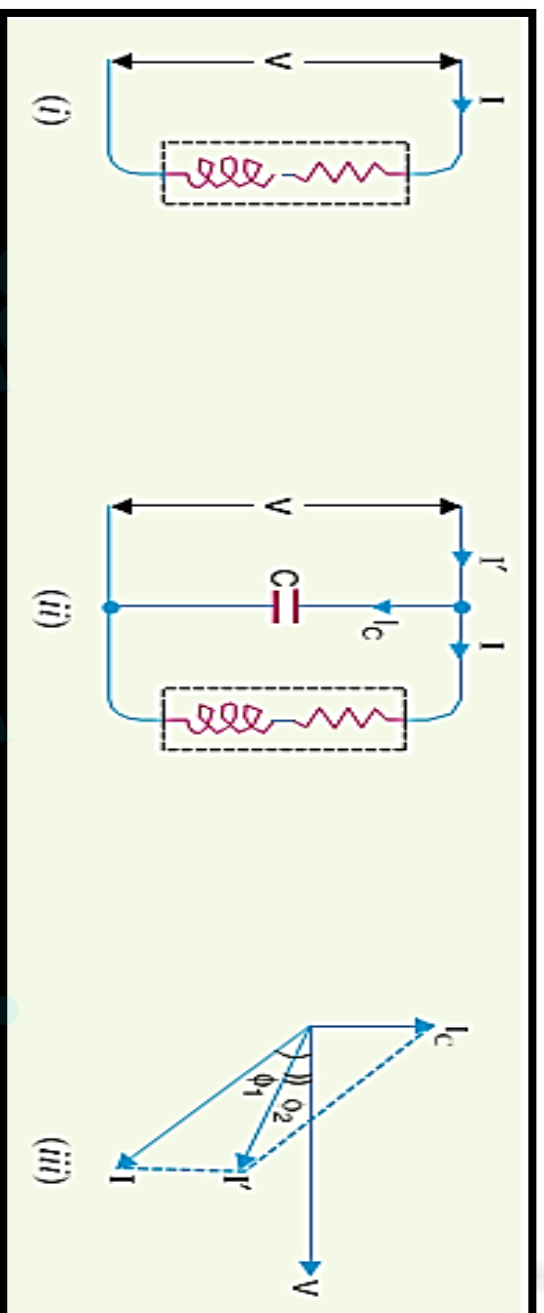
5. Reduced handling capacity of system. The lagging power factor reduces the handling capacity of all the elements of the system. It is because the reactive component of current prevents the full utilisation of installed capacity.

Causes of Low Power Factor

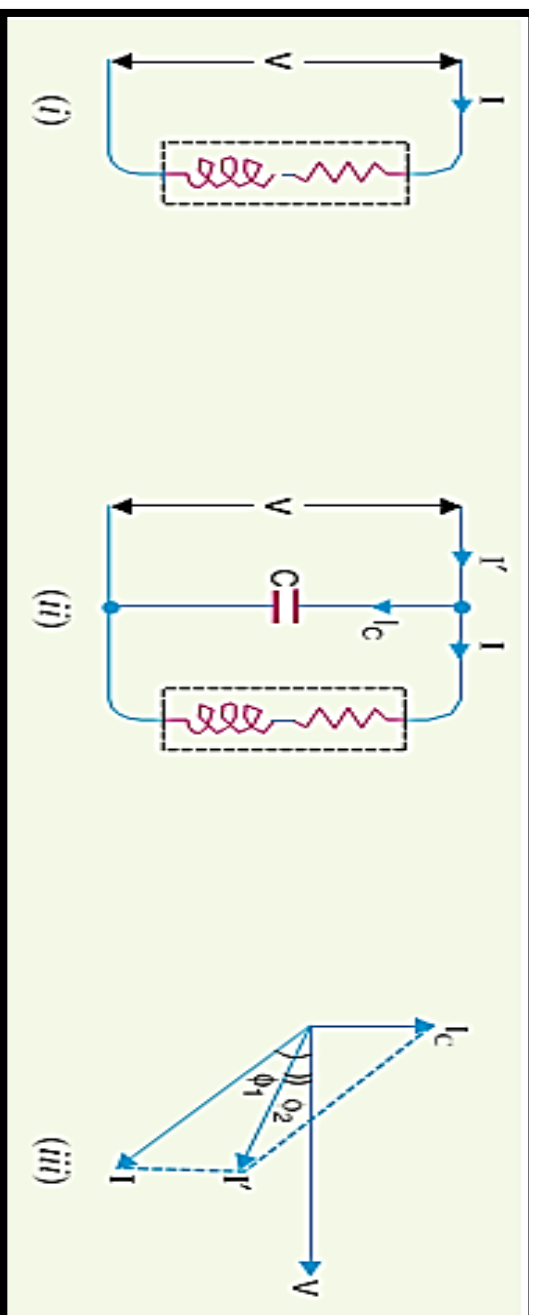
- Low power factor is undesirable from economic point of view. Normally, the power factor of the whole load on the supply system is lower than 0.8. The following are the causes of low power factor:
 1. Most of the a.c. motors are of induction type (1 ϕ and 3 ϕ induction motors) which have low lagging power factor. These motors work at a power factor which is extremely small on light load (0.2 to 0.3) and rises to 0.8 or 0.9 at full load.
 2. Arc lamps, electric discharge lamps and industrial heating furnaces operate at low lagging power factor.
 3. The load on the power system is varying ; being high during morning and evening and low at other times. During low load period, supply voltage is increased which increases the magnetisation current. This results in the decreased power factor.



Power Factor Improvement

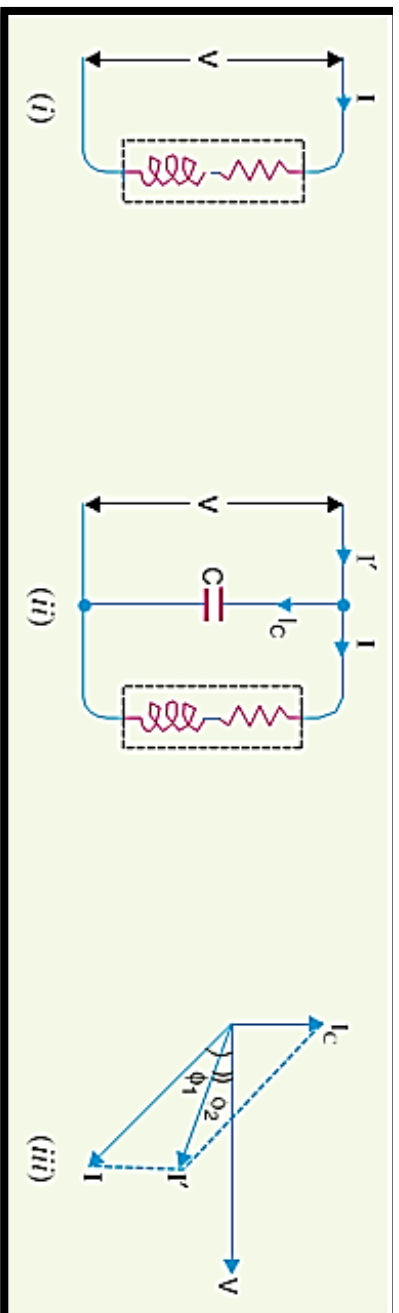


- The low power factor is mainly due to the fact that most of the power loads are inductive and, therefore, take lagging currents.
- In order to improve the power factor, some device taking leading power should be connected in parallel with the load.
- One of such devices can be a capacitor.
- The capacitor draws a leading current and partly or completely neutralises the lagging reactive component of load current.
- This raises the power factor of the load.



Illustration

- To illustrate the power factor improvement by a capacitor, consider a single phase load taking lagging current I at a power factor $\cos \phi_1$ as shown in Fig.
- The capacitor C is connected in parallel with the load.
- The capacitor draws current I_C which leads the supply voltage by 90° .
- The resulting line current I' is the phasor sum of I and I_C and its angle of lag is ϕ_2 as shown in the phasor diagram of Fig.
- It is clear that ϕ_2 is less than ϕ_1 , so that $\cos \phi_2$ is greater than $\cos \phi_1$. Hence, the power factor of the load is improved.



1. The circuit current I' after p.f. correction is less than the original circuit current I .
2. The active or wattful component remains the same before and after p.f. correction because only the lagging reactive component is reduced by the capacitor.

$$I \cos \phi_1 = I' \cos \phi_2$$

3. The lagging reactive component is reduced after p.f. improvement and is equal to the difference between lagging reactive component of load ($I \sin \phi_1$) and capacitor current (I_c).

$$I' \sin \phi_2 = I \sin \phi_1$$

4. As

$$I \cos \phi_1 = I' \cos \phi_2$$

$$VI \cos \phi_1 = V I' \cos \phi_2$$

Therefore, **active power (kW) remains unchanged due to power factor improvement.**

5. As

$$I' \sin \phi_2 = I \sin \phi_1$$

$$V I' \sin \phi_2 = V I \sin \phi_1$$

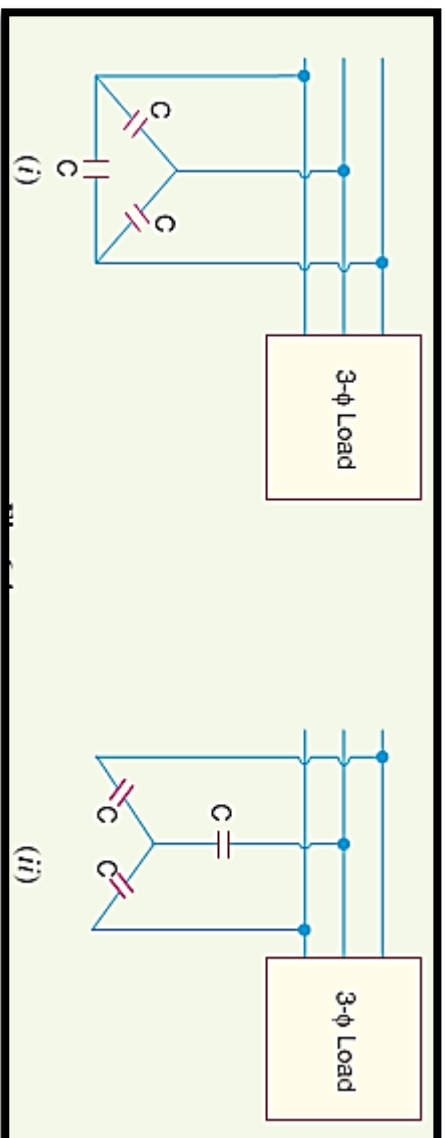
Net kVAR after p.f. correction = Lagging kVAR before p.f. correction - leading kVAR of equipment

Power Factor Improvement Equipment

- Normally, the power factor of the whole load on a large generating station is in the region of 0.8 to 0.9.
- However, sometimes it is lower and in such cases it is generally desirable to take special steps to improve the power factor.
 1. Static capacitors.
 2. Synchronous condenser.
 3. Phase advancers



1. Static capacitor



- The power factor can be improved by connecting capacitors in parallel with the equipment operating at lagging power factor.
- The capacitor (generally known as static capacitor) draws a leading current and partly or completely neutralises the lagging reactive component of load current.
- This raises the power factor of the load. For three-phase loads, the capacitors can be connected in delta or star as shown in Fig.
- Static capacitors are invariably used for power factor improvement in factories.

Advantages

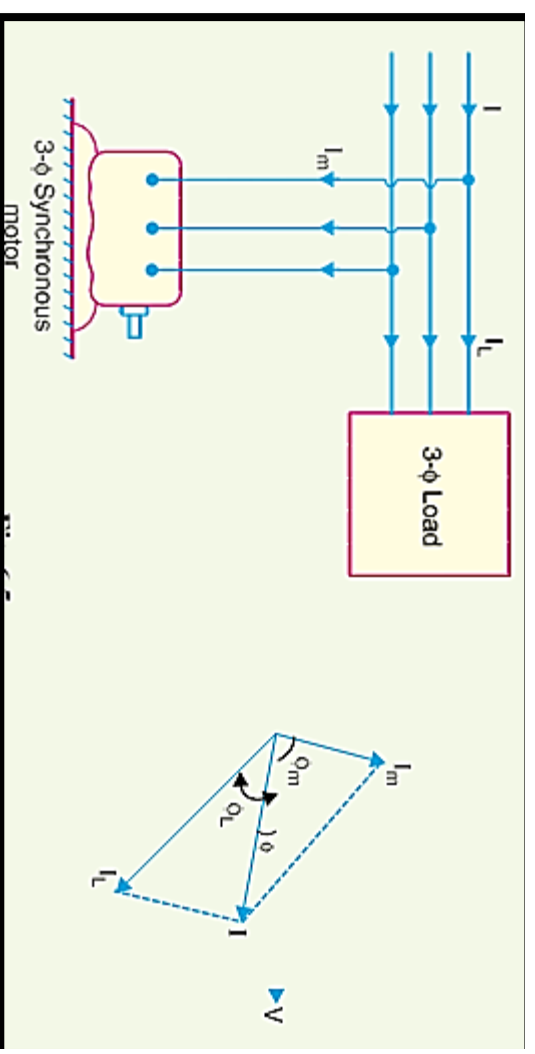
- They have low losses.
- They require little maintenance as there are no rotating parts.
- They can be easily installed as they are light and require no foundation.
- They can work under ordinary atmospheric conditions.

Disadvantages

- They have short service life ranging from 8 to 10 years.
- They are easily damaged if the voltage exceeds the rated value.
- Once the capacitors are damaged, their repair is uneconomical.



2. Synchronous Condenser



A synchronous motor takes a leading current when over-excited and, therefore, behaves as a capacitor.

An over-excited synchronous motor running on no load is known as *synchronous condenser*.

When such a machine is connected in parallel with the supply, it takes a leading current which partly neutralises the lagging reactive component of the load.

Thus the power factor is improved.

Fig shows the power factor improvement by synchronous condenser method.

The 3 ϕ load takes current I_L at low lagging power factor $\cos \phi_L$.

The synchronous condenser takes a current I_m which leads the voltage by an angle ϕ_m .

The resultant current I is the phasor sum of I_m and I_L and lags behind the voltage by an angle ϕ .

It is clear that ϕ is less than ϕ_L so that $\cos \phi$ is greater than $\cos \phi_L$.

Thus the power factor is increased from $\cos \phi_L$ to $\cos \phi$.

Synchronous condensers are generally used at major bulk supply substations for power factor improvement

Advantages

- By varying the field excitation, the magnitude of current drawn by the motor can be changed by any amount. This helps in achieving stepless control of power factor.
- The motor windings have high thermal stability to short circuit currents.
- The faults can be removed easily.

Disadvantages

- There are considerable losses in the motor.
- The maintenance cost is high.
- It produces noise.
- Except in sizes above 500 kVA, the cost is greater than that of static capacitors of the same rating.
- As a synchronous motor has no self-starting torque, therefore, an auxiliary equipment has to be provided for this purpose.



3. Phase Advancers

- Phase advancers are used to improve the power factor of induction motors.
- The low power factor of an induction motor is due to the fact that its stator winding draws exciting current which lags behind the supply voltage by 90° .
- If the exciting ampere turns can be provided from some other a.c. source, then the stator winding will be relieved of exciting current and the power factor of the motor can be improved.
- This job is accomplished by the phase advancer which is simply an a.c. exciter
- The phase advancer is mounted on the same shaft as the main motor and is connected in the rotor circuit of the motor. It provides exciting ampere turns to the rotor circuit at slip frequency.
- By providing more ampere turns than required, the induction motor can be made to operate on leading power factor like an over-excited synchronous motor.

Advantages

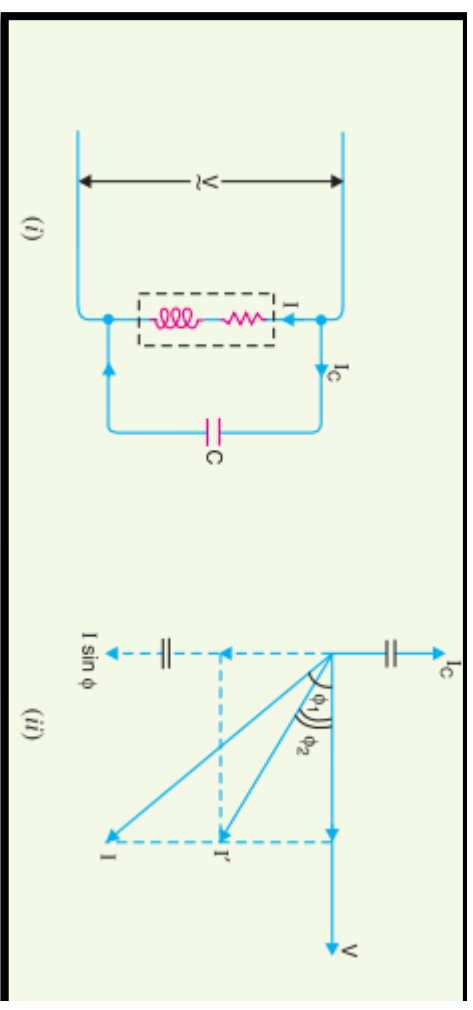
- As the exciting ampere turns are supplied at slip frequency, therefore, lagging kVAR drawn by the motor are considerably reduced.
- Phase advancer can be conveniently used where the use of synchronous motors is inadmissible.

Disadvantage

- They are not economical



Power factor correction calculations



From the phasor diagram, it is clear that after p.f. correction, the lagging reactive component of the load is reduced to $I' \sin \phi_2$,

Obviously,

$$I' \sin \phi_2 = I \sin \phi_1 - I_C$$

or

$$I_C = I \sin \phi_1 - I' \sin \phi_2$$

$\therefore$ Capacitance of capacitor to improve p.f. from $\cos \phi_1$ to $\cos \phi_2$

$$= \frac{I_C}{\omega V}$$

$$\left(\because X_C = \frac{V}{I_C} = \frac{1}{\omega C} \right)$$

Example 3.1. The maximum demand on a power station is 100 MW. If the annual load factor is 40%, calculate the total energy generated in a year.

Solution.

$$\begin{aligned}\text{Energy generated/year} &= \text{Max. demand} \times \text{L.F.} \times \text{Hours in a year} \\ &= (100 \times 10^3) \times (0.4) \times (24 \times 365) \text{ kWh} \\ &= \mathbf{3504 \times 10^5 \text{ kWh}}\end{aligned}$$

Example 3.2. A generating station has a connected load of 43MW and a maximum demand of 20 MW; the units generated being 61.5×10^6 per annum. Calculate (i) the demand factor and (ii) load factor.

Solution.

$$(i) \quad \text{Demand factor} = \frac{\text{Max. demand}}{\text{Connected load}} = \frac{20}{43} = \mathbf{0.465}$$

$$(ii) \quad \text{Average demand} = \frac{\text{Units generated / annum}}{\text{Hours in a year}} = \frac{61.5 \times 10^6}{8760} = 7020 \text{ kW}$$

$$\therefore \quad \text{Load factor} = \frac{\text{Average demand}}{\text{Max. demand}} = \frac{7020}{20 \times 10^3} = \mathbf{0.351 \text{ or } 35.1\%}$$

Example 3.3. A 100 MW power station delivers 100 MW for 2 hours, 50 MW for 6 hours and is shut down for the rest of each day. It is also shut down for maintenance for 45 days each year. Calculate its annual load factor.

Solution.

Energy supplied for each working day

$$= (100 \times 2) + (50 \times 6) = 500 \text{ MWh}$$

$$\text{Station operates for} = 365 - 45 = 320 \text{ days in a year}$$

$$\therefore \quad \text{Energy supplied/year} = 500 \times 320 = 160,000 \text{ MWh}$$

$$\begin{aligned}\text{Annual load factor} &= \frac{\text{MWh supplied per annum}}{\text{Max. demand in MW} \times \text{Working hours}} \times 100 \\ &= \frac{160,000}{(100) \times (320 \times 24)} \times 100 = \mathbf{20.8\%}\end{aligned}$$

Example 3.4. A generating station has a maximum demand of 25MW, a load factor of 60%, a plant capacity factor of 50% and a plant use factor of 72%. Find (i) the reserve capacity of the plant (ii) the daily energy produced and (iii) maximum energy that could be produced daily if the plant while running as per schedule, were fully loaded.

Solution.

$$(i) \quad \text{Load factor} = \frac{\text{Average demand}}{\text{Maximum demand}}$$

$$\text{or} \quad 0.60 = \frac{\text{Average demand}}{25}$$

$$\therefore \quad \text{Average demand} = 25 \times 0.60 = 15 \text{ MW}$$

$$\text{Plant capacity factor} = \frac{\text{Average demand}}{\text{Plant capacity}}$$

$$\therefore \quad \text{Plant capacity} = \frac{\text{Average demand}}{\text{Plant capacity factor}} = \frac{15}{0.5} = 30 \text{ MW}$$

$$\therefore \quad \text{Reserve capacity of plant} = \text{Plant capacity} - \text{maximum demand} \\ = 30 - 25 = \mathbf{5 \text{ MW}}$$

$$(ii) \quad \text{Daily energy produced} = \text{Average demand} \times 24 \\ = 15 \times 24 = \mathbf{360 \text{ MWh}}$$

$$(iii) \quad \text{Maximum energy that could be produced} \\ = \frac{\text{Actual energy produced in a day}}{\text{Plant use factor}} \\ = \frac{360}{0.72} = \mathbf{500 \text{ MWh/day}}$$

Example 3.5. A diesel station supplies the following loads to various consumers :

Industrial consumer = 1500 kW ; Commercial establishment = 750 kW

Domestic power = 100 kW; Domestic light = 450 kW

If the maximum demand on the station is 2500 kW and the number of kWh generated per year is 5×10^5 , determine (i) the diversity factor and (ii) annual load factor.

Solution.

$$(i) \quad \text{Diversity factor} = \frac{1500 + 750 + 100 + 450}{2500} = \mathbf{1.12}$$

$$(ii) \quad \text{Average demand} = \frac{\text{kWh generated / annum}}{\text{Hours in a year}} = 45 \times 10^5 / 8760 = 513.7 \text{ kW}$$

$$\therefore \quad \text{Load factor} = \frac{\text{Average load}}{\text{Max. demand}} = \frac{513.7}{2500} = 0.205 = \mathbf{20.5\%}$$

Example 3.6. A power station has a maximum demand of 15000 kW. The annual load factor is 40% and plant capacity factor is 40%. Determine the reserve capacity of the plant.

Solution.

$$\begin{aligned} \text{Energy generated/annum} &= \text{Max. demand} \times \text{L.F.} \times \text{Hours in a year} \\ &= (15000) \times (0.5) \times (8760) \text{ kWh} \\ &= 65.7 \times 10^6 \text{ kWh} \end{aligned}$$

$$\text{Plant capacity factor} = \frac{\text{Units generated / annum}}{\text{Plant capacity} \times \text{Hours in a year}}$$

$$\therefore \quad \text{Plant capacity} = \frac{65.7 \times 10^6}{0.4 \times 8760} = 18,750 \text{ kW}$$

$$\text{Reserve capacity} = \text{Plant capacity} - \text{Max. demand}$$

Example 3.10. A generating station has the following daily load cycle :

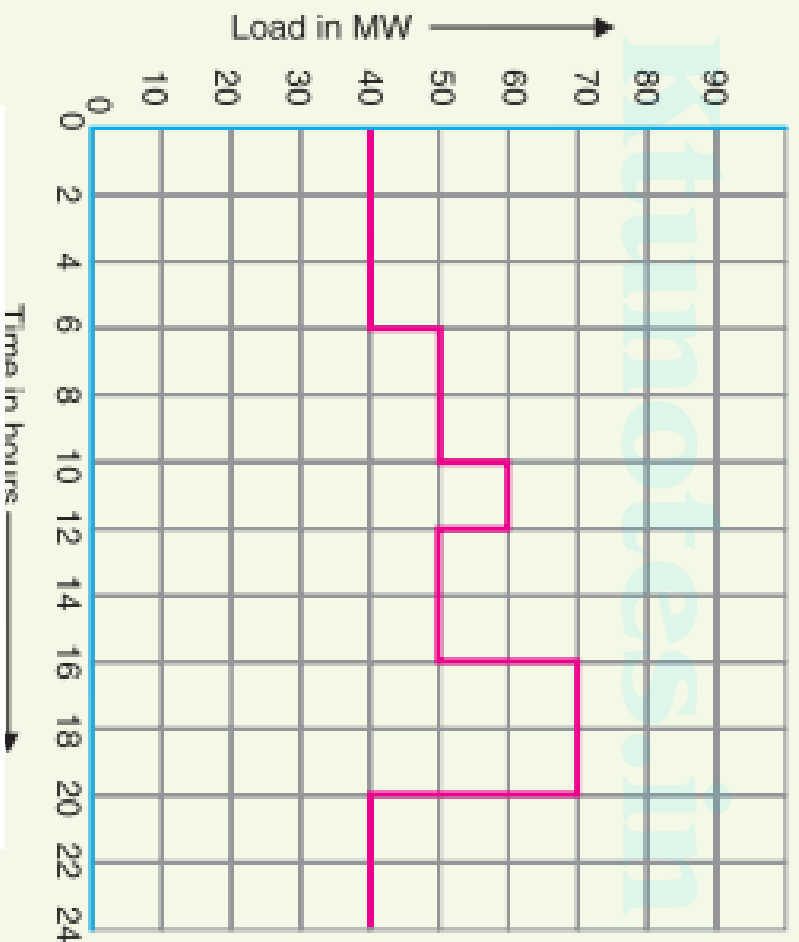
Time (Hours)	0—6	6—10	10—12	12—16	16—20	20—24
Load (MW)	40	50	60	50	70	40

Draw the load curve and find (i) maximum demand (ii) units generated per day (iii) average load and (iv) load factor.

Solution. Daily curve is drawn by taking the load along Y-axis and time along X-axis. For the given load cycle, the load curve is shown in Fig. 3.6.

(i) It is clear from the load curve that maximum demand on the power station is 70 MW and occurs during the period 16—20 hours.

$$\therefore \text{Maximum demand} = \mathbf{70 \text{ MW}}$$



(ii) Units generated/day = Area (in kWh) under the load curve

$$\begin{aligned} &= 10^3 [40 \times 6 + 50 \times 4 + 60 \times 2 + 50 \times 4 + 70 \times 4 + 40 \times 4] \\ &= 10^3 [240 + 200 + 120 + 200 + 280 + 160] \text{ kWh} \\ &= \mathbf{12 \times 10^5 \text{ kWh}} \end{aligned}$$

$$(iii) \quad \text{Average load} = \frac{\text{Units generated / day}}{24 \text{ hours}} = \frac{12 \times 10^5}{24} = \mathbf{50,000 \text{ kW}}$$

$$(iv) \quad \text{Load factor} = \frac{\text{Average load}}{\text{Max. demand}} = \frac{50,000}{70 \times 10^3} = 0.714 = \mathbf{71.4\%}$$

Example 3.11. A power station has to meet the following demand :

Group A : 200 kW between 8 A.M. and 6 P.M.

Group B : 100 kW between 6 A.M. and 10 A.M.

Group C : 50 kW between 6 A.M. and 10 A.M.

Group D : 100 kW between 10 A.M. and 6 P.M. and then between 6 P.M. and 6 A.M.

Plot the daily load curve and determine (i) diversity factor (ii) units generated per day (iii) load factor.

Solution. The given load cycle can be tabulated as under :

Time (Hours)	0—6	6—8	8—10	10—18	18—24
Group A	—	—	200 kW	200 kW	—
Group B	—	100 kW	100 kW	—	—
Group C	—	50 kW	50 kW	—	—
Group D	100 kW	—	—	100 kW	100 kW
Total load on power station	100 kW	150 kW	250 kW	300 kW	100 kW

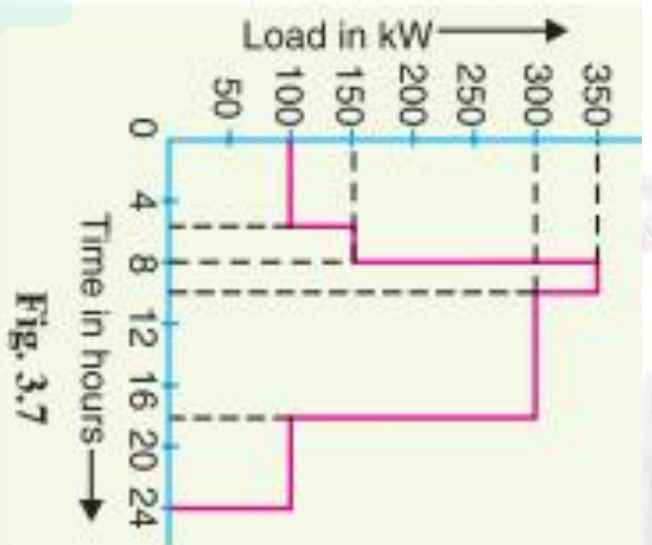
From this table, it is clear that total load on power station is 100 kW for 0—6 hours, 150 kW for 6—8 hours, 350 kW for 8—10 hours, 300 kW for 10—18 hours and 100 kW for 18—24 hours. Plotting the load on power station versus time, we get the daily load curve as shown in Fig. 3.7. It is clear from the curve that maximum demand on the station is 350 kW and occurs from 8 A.M. to 10 A.M. i.e.,

$$\text{Maximum demand} = 350 \text{ kW}$$

Sum of individual maximum demands of groups

$$= 200 + 100 + 50 + 100 \\ = 450 \text{ kW}$$

Fig. 3.7



$$(i) \quad \text{Diversity factor} = \frac{\text{Sum of individual max. demands}}{\text{Max. demand on station}} = 450/350 = \mathbf{1.286}$$

$$(ii) \quad \begin{aligned} \text{Units generated/day} &= \text{Area (in kWh) under load curve} \\ &= 100 \times 6 + 150 \times 2 + 350 \times 2 + 300 \times 8 + 100 \times 6 \\ &= \mathbf{4600 \text{ kWh}} \end{aligned}$$

$$(iii) \quad \text{Average load} = 4600/24 = 191.7 \text{ kW}$$

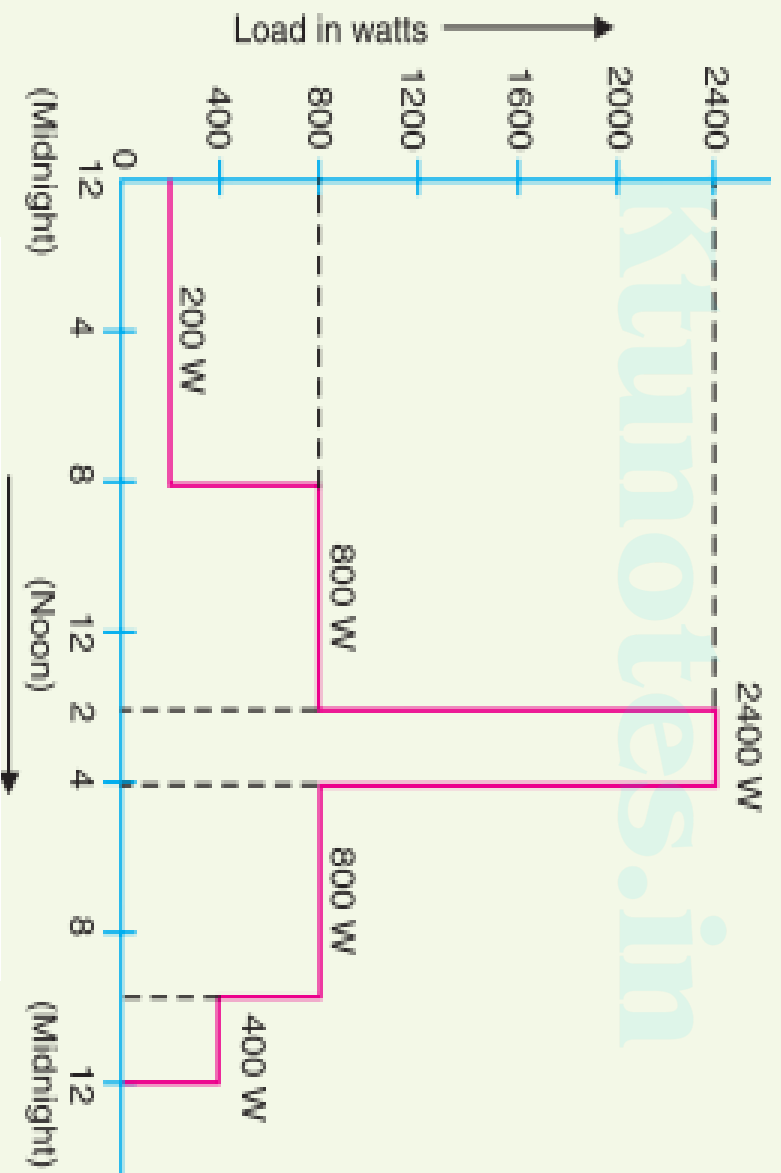
$$\therefore \quad \text{Load factor} = \frac{191.7}{350} \times 100 = \mathbf{54.8\%}$$

Example 3.12. The daily demands of three consumers are given below :

Time	Consumer 1	Consumer 2	Consumer 3
12 midnight to 8 A.M.	No load	200 W	No load
8 A.M. to 2 P.M.	600 W	No load	200 W
2 P.M. to 4 P.M.	200 W	1000 W	1200 W
4 P.M. to 10 P.M.	800 W	No load	No load
10 P.M. to midnight	No load	200 W	200 W

Plot the load curve and find (i) maximum demand of individual consumer (ii) load factor of individual consumer (iii) diversity factor and (iv) load factor of the station.

Solution. Fig. 3.8 shows the load curve.



(i) Max. demand of consumer 1 = **800 W**

Max. demand of consumer 2 = **1000 W**

Max. demand of consumer 3 = **1200 W**

(ii)

$$\text{L.F. of consumer 1} = \frac{\text{Energy consumed / day}}{\text{Max. demand} \times \text{Hours in a day}} \times 100$$

$$= \frac{600 \times 6 + 200 \times 2 + 800 \times 6}{800 \times 24} \times 100 = \mathbf{45.8\%}$$

$$\text{L.F. of consumer 2} = \frac{200 \times 8 + 1000 \times 2 + 200 \times 2}{1000 \times 24} \times 100 = \mathbf{16.7\%}$$

$$\text{L.F. of consumer 3} = \frac{200 \times 6 + 1200 \times 2 + 200 \times 2}{1200 \times 24} \times 100 = \mathbf{13.8\%}$$

(iii) The simultaneous maximum demand on the station is $200 + 1000 + 1200 = 2400 \text{ W}$ and occurs from 2 P.M. to 4 P.M.

$$\therefore \text{Diversity factor} = \frac{800 + 1000 + 1200}{2400} = \mathbf{1.25}$$

(iv)

$$\begin{aligned} \text{Station load factor} &= \frac{\text{Total energy consumed / day}}{\text{Simultaneous max. demand} \times 24} \times 100 \\ &= \frac{8800 + 4000 + 4000}{2400 \times 24} \times 100 = \mathbf{29.1\%} \end{aligned}$$

Example 3.15. A power station has the following daily load cycle :

Time in Hours	6—8	8—12	12—16	16—20	20—24	24—6
Load in MW	20	40	60	20	50	20

Plot the load curve and load duration curve. Also calculate the energy generated per day.

Solution. Fig. 3.9 (i) shows the daily load curve, whereas Fig. 3.9 (ii) shows the daily load duration curve. It can be readily seen that area under the two load curves is the same. Note that load duration curve is drawn by arranging the loads in the order of descending magnitudes.

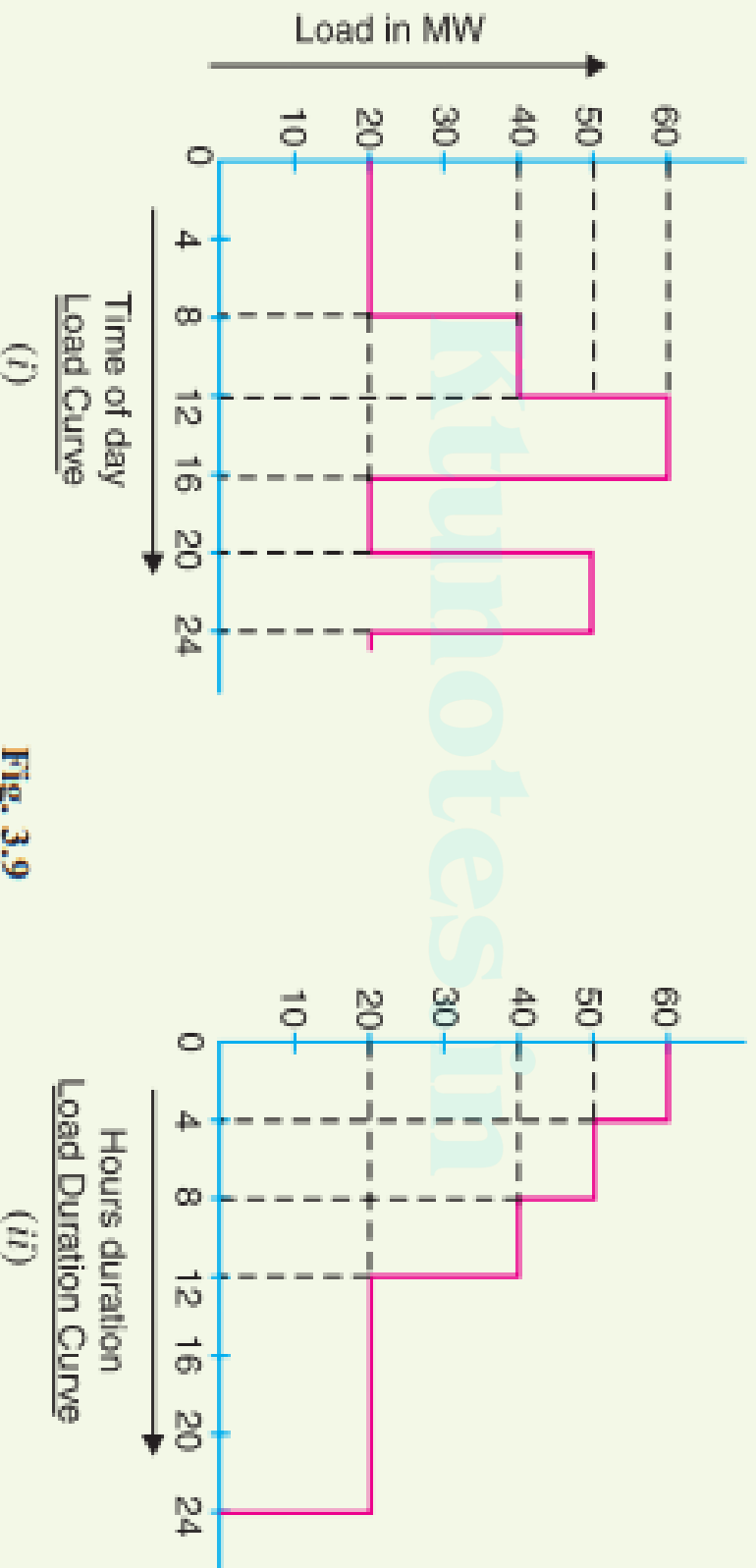


Fig. 3.9

$$\begin{aligned}
 \text{Units generated/day} &= \text{Area (in kWh) under daily load curve} \\
 &= 10^3 [20 \times 8 + 40 \times 4 + 60 \times 4 + 20 \times 4 + 50 \times 4]
 \end{aligned}$$

Example 3.18. A proposed station has the following daily load cycle :

Time in hours	6—8	8—11	11—16	16—19	19—22	22—24	24—6
Load in MW	20	40	50	35	70	40	20

Draw the load curve and select suitable generator units from the 10,000, 20,000, 25,000, 30,000 kVA. Prepare the operation schedule for the machines selected and determine the load factor from the curve

Solution. The load curve of the power station can be drawn to some suitable scale as shown in Fig. 3.12.

Units generated per day = Area (in kWh) under the load curve

$$\begin{aligned}
 &= 10^3 [20 \times 8 + 40 \times 3 + 50 \times 5 + 35 \times 3 + 70 \times 3 + 40 \times 2] \\
 &= 10^3 [160 + 120 + 250 + 105 + 210 + 80] \text{ kWh} \\
 &= 925 \times 10^3 \text{ kWh}
 \end{aligned}$$

$$\text{Average load} = \frac{925 \times 10^3}{24} = 38541.7 \text{ kW}$$

$$\text{Load factor} = \frac{38541.7}{70 \times 10^3} \times 100 = 55.06\%$$

Selection of number and sizes of units : Assuming power factor of the machines to be 0.8, the output of the generating units available will be 8, 16, 20 and 24 MW. There can be several possibilities. However, while selecting the size and number of units, it has to be borne in mind that (i) one set of highest capacity should be kept as standby unit (ii) the units should meet the maximum demand (70 MW in this case) on the station (iii) there should be overall economy.

Keeping in view the above facts, 4 sets of 24 MW each may be chosen. Three sets will meet the maximum demand of 70 MW and one unit will serve as a standby unit.

Operational schedule. Referring to the load curve shown in Fig. 3.12, the operational schedule will be as under :

- (i) Set No. 1 will run for 24 hours.
- (ii) Set No. 2 will run from 8.00 hours to midnight.
- (iii) Set No. 3 will run from

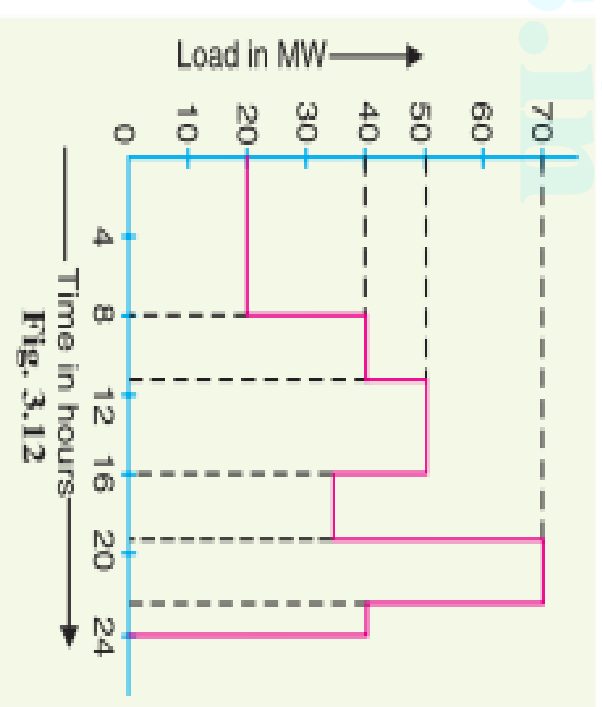


Fig. 3.12

from 19 hours to 22 hours.

Power factor improvement

Example 6.1 *An alternator is supplying a load of 300 kW at a p.f. of 0.6 lagging. If the power factor is raised to unity, how many more kilowatts can alternator supply for the same kVA loading ?*

Solution :

$$\text{kVA} = \frac{\text{kW}}{\cos \phi} = \frac{300}{0.6} = 500 \text{ kVA}$$

$$\text{kW at 0.6 p.f.} = 300 \text{ kW}$$

$$\text{kW at 1 p.f.} = 500 \times 1 = 500 \text{ kW}$$

∴ Increased power supplied by the alternator

$$= 500 - 300 = \mathbf{200 \text{ kW}}$$

Note the importance of power factor improvement. When the p.f. of the alternator is unity, the 500 kVA are also 500 kW and the engine driving the alternator has to be capable of developing this power together with the losses in the alternator. But when the power factor of the load is 0.6, the power is only 300 kW. Therefore, the engine is developing only 300 kW, though the alternator is supplying its rated output of 500 kVA.

Example 6.2 A single phase motor connected to 400 V, 50 Hz supply takes 31.7 A at a power factor of 0.7 lagging. Calculate the capacitance required in parallel with the motor to raise the power factor to 0.9 lagging.

Solution : The circuit and phasor diagrams are shown in Figs. 6.8 and 6.9 respectively. Here motor M is taking a current I_M of 31.7 A. The current I_C taken by the capacitor must be such that when combined with I_M , the resultant current I lags the voltage by an angle ϕ where $\cos \phi = 0.9$.

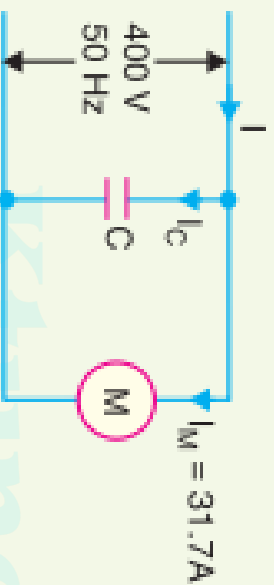


Fig. 6.8

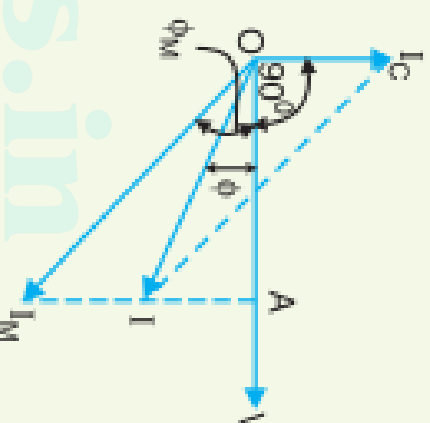


Fig. 6.9

Referring to the phasor diagram in Fig. 6.9,

$$\text{Active component of } I_M = I_M \cos \phi_M = 31.7 \times 0.7 = 22.19 \text{ A}$$

$$\text{Active component of } I = I \cos \phi = I \times 0.9$$

These components are represented by OA in Fig. 6.9.

$$\therefore I = \frac{22.19}{0.9} = 24.65 \text{ A}$$

$$\text{Reactive component of } I_M = I_M \sin \phi_M = 31.7 \times 0.714^* = 22.6 \text{ A}$$

$$\begin{aligned} \text{Reactive component of } I &= I \sin \phi = 24.65 \sqrt{1 - (0.9)^2} \\ &= 24.65 \times 0.436 = 10.75 \text{ A} \end{aligned}$$

It is clear from Fig. 6.9 that :

$$I_C = \text{Reactive component of } I_M - \text{Reactive component of } I \\ = 22.6 - 10.75 = 11.85 \text{ A}$$

But
$$I_C = \frac{V}{X_C} = V \times 2\pi f C$$

or
$$11.85 = 400 \times 2\pi \times 50 \times C$$

$\therefore C = 94.3 \times 10^{-6} \text{ F} = \mathbf{94.3 \mu F}$

Note the effect of connecting a $94.3 \mu\text{F}$ capacitor in parallel with the motor. The current taken from the supply is reduced from 31.7 A to 24.65 A without altering the current or power taken by the motor. This enables an economy to be affected in the size of generating plant and in the cross-sectional area of the conductors.

Example 6.4 A 3-phase, 5 kW induction motor has a p.f. of 0.75 lagging. A bank of capacitors is connected in delta across the supply terminals and p.f. raised to 0.9 lagging. Determine the kVAR rating of the capacitors connected in each phase.

Solution :

Original p.f., $\cos \phi_1 = 0.75$ lag ; Motor input, $P = 5 \text{ kW}$

Final p.f., $\cos \phi_2 = 0.9$ lag ; Efficiency, $\eta = 100\%$ (assumed)

$\phi_1 = \cos^{-1}(0.75) = 41.41^\circ$; $\tan \phi_1 = \tan 41.41^\circ = 0.8819$

$\phi_2 = \cos^{-1}(0.9) = 25.84^\circ$; $\tan \phi_2 = \tan 25.84^\circ = 0.4843$

Leading kVAR taken by the condenser bank

$$= P (\tan \phi_1 - \tan \phi_2) \\ = 5 (0.8819 - 0.4843) = 1.99 \text{ kVAR}$$

$\therefore$ Rating of capacitors connected in each phase